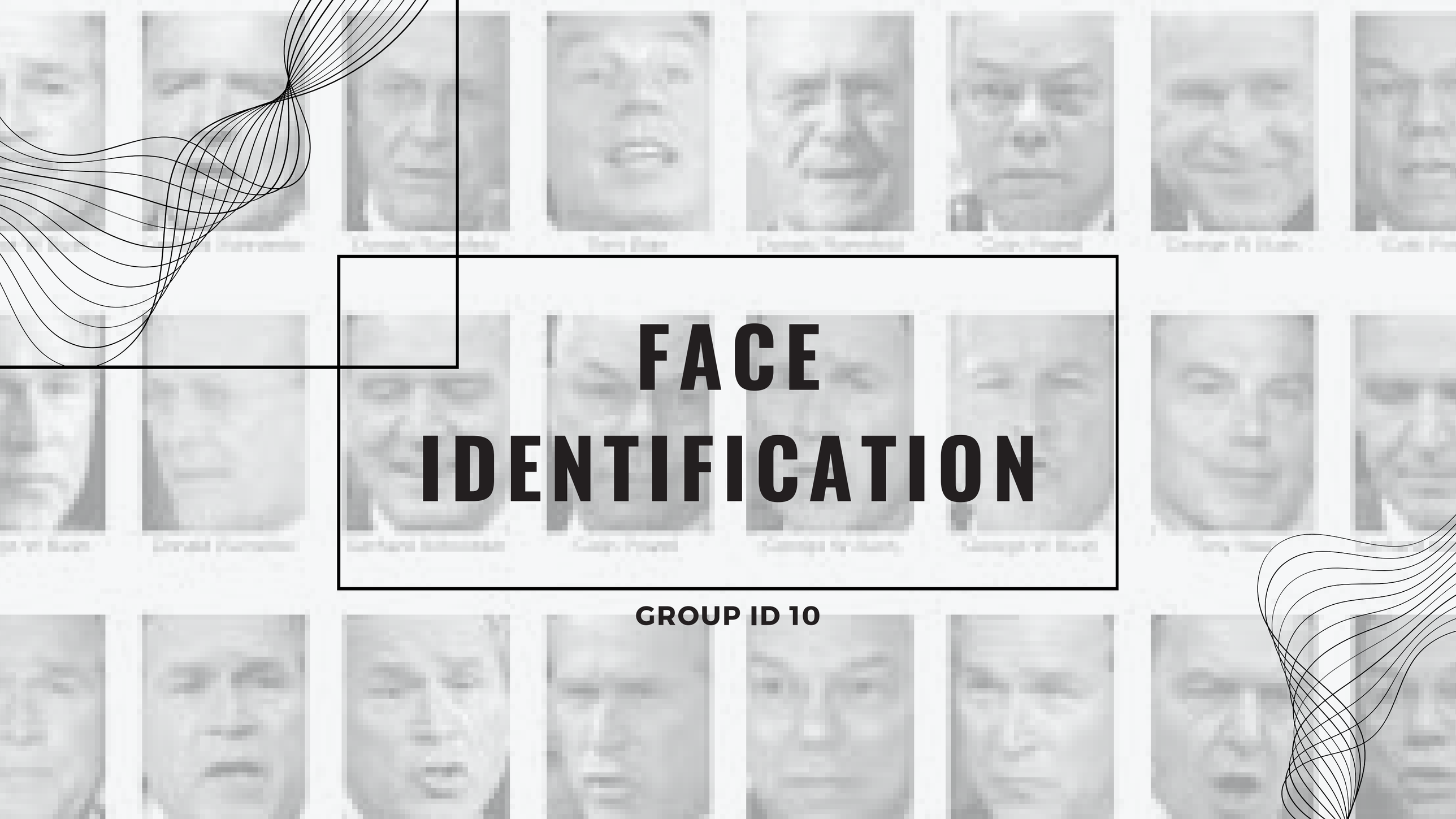


FACE IDENTIFICATION

GROUP ID 10

CONTENT

01 DATA ANALYSIS

02 PCA ON ORIGINAL DATA

03 LDA ON PCA DATA

04 HOG FEATURE EXTRACTION

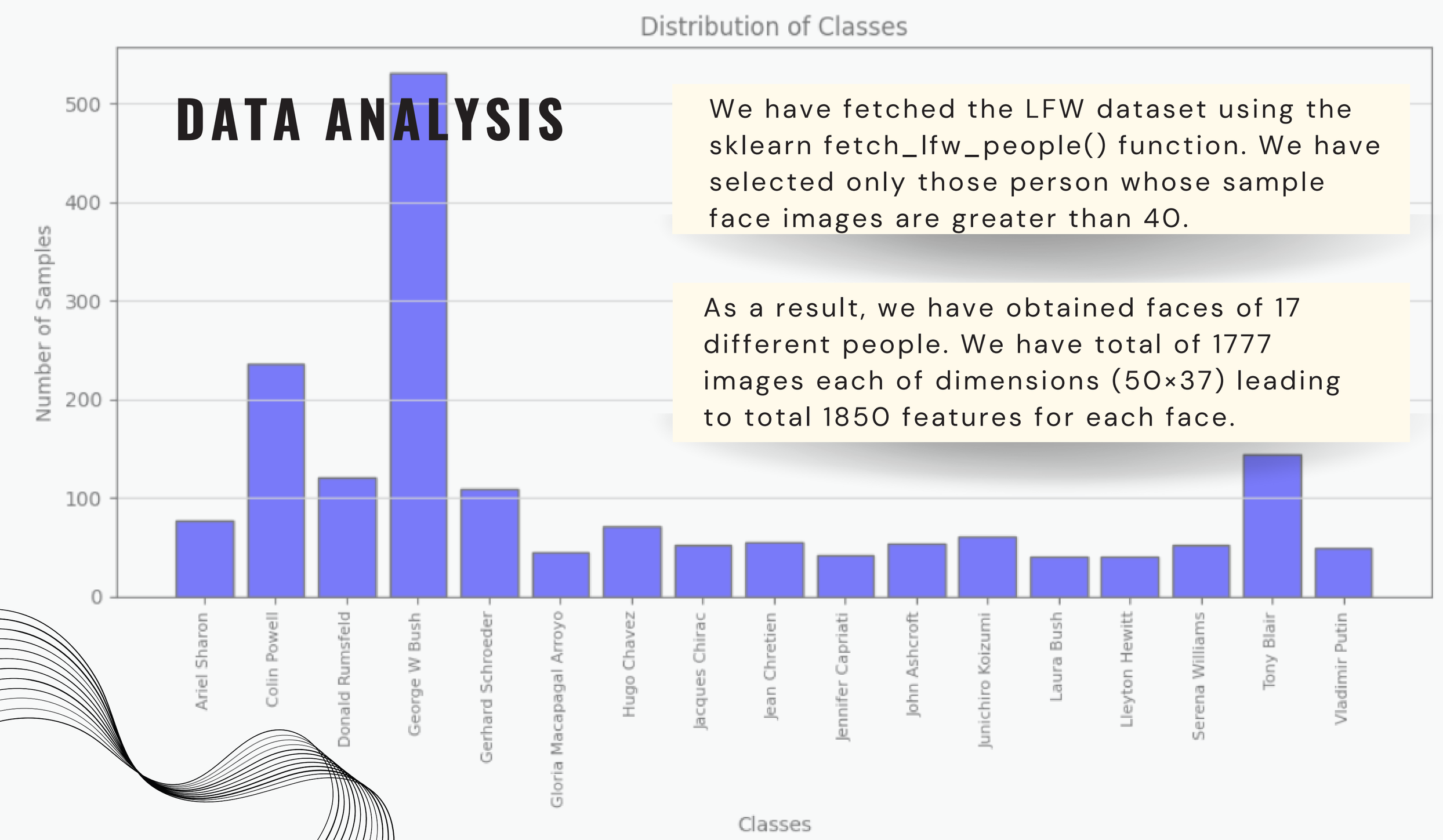
05 LBP FEATURE EXTRACTION

06 CNN FEATURE EXTRACTION

DATA ANALYSIS

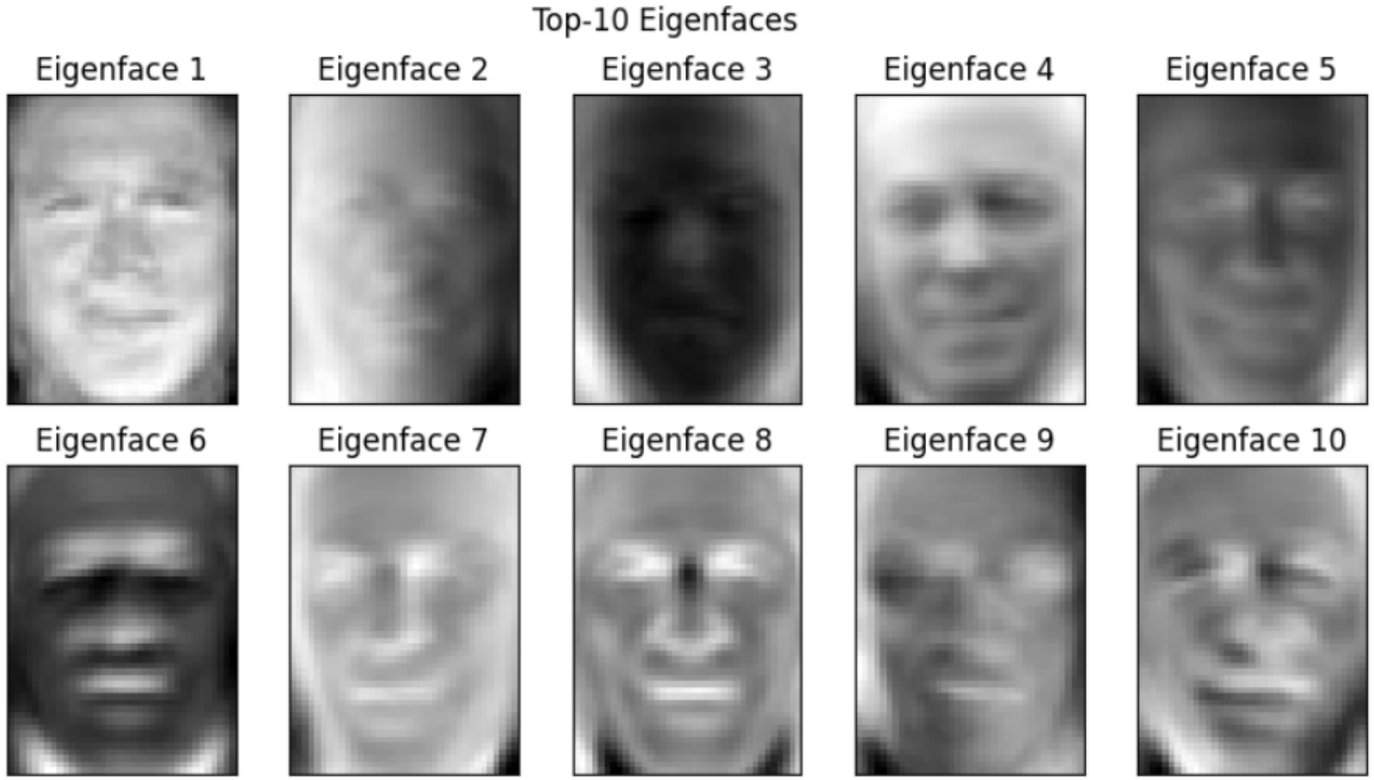
We have fetched the LFW dataset using the sklearn fetch_lfw_people() function. We have selected only those person whose sample face images are greater than 40.

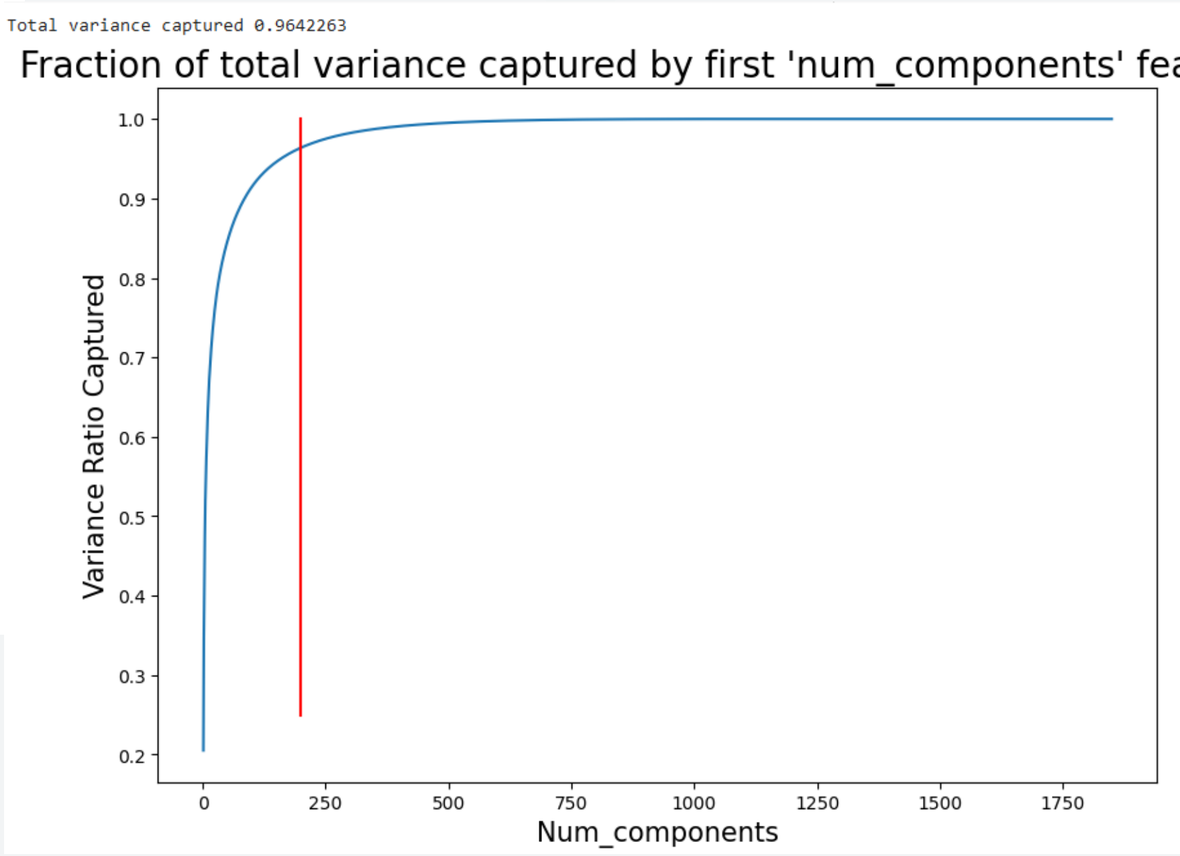
As a result, we have obtained faces of 17 different people. We have total of 1777 images each of dimensions (50×37) leading to total 1850 features for each face.



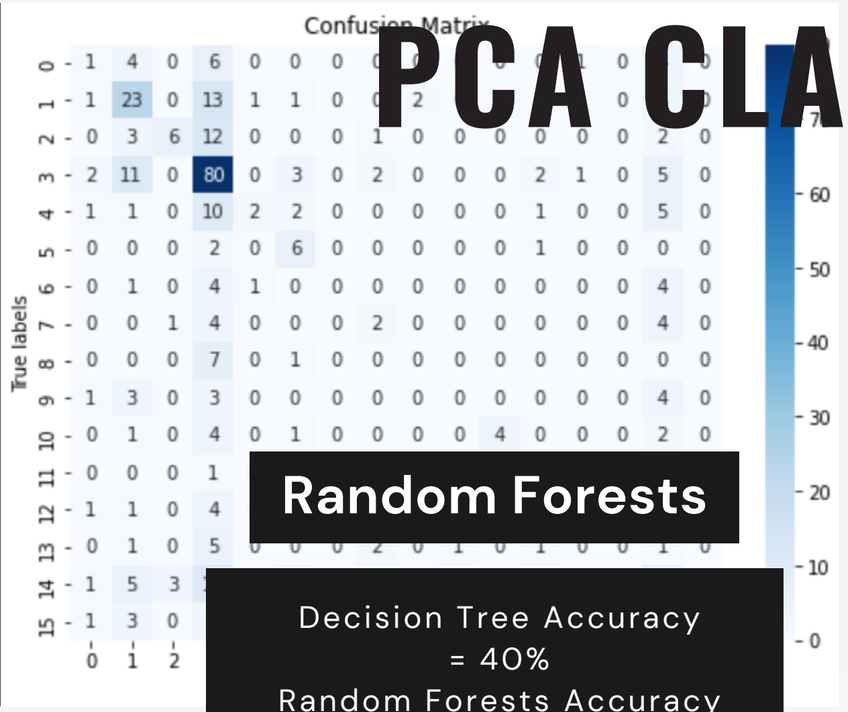
PRINCIPAL COMPONENT ANALYSIS

Applying principal component analysis, we have reduced 1850 features of the training data to 200 features still capturing 96.4% of the total variance.



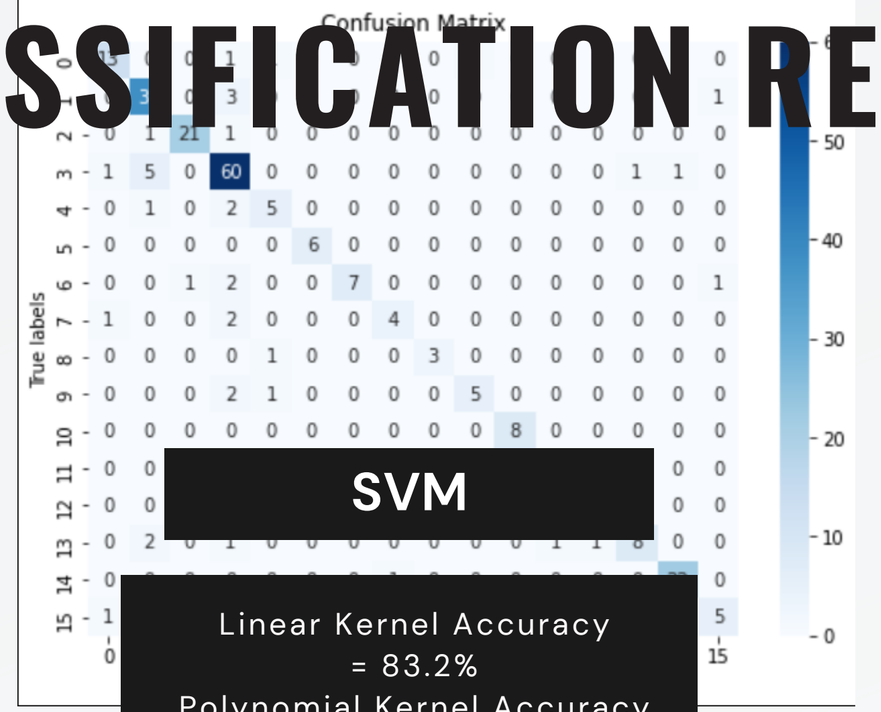


PCA CLASSIFICATION RESULTS



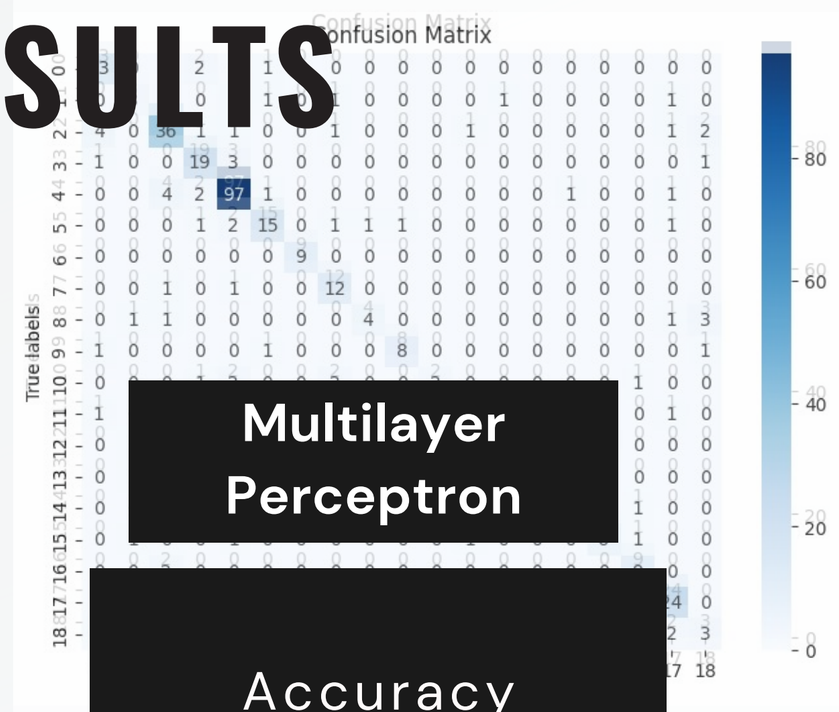
Random Forests

Decision Tree Accuracy
= 40%
Random Forests Accuracy
= 40%



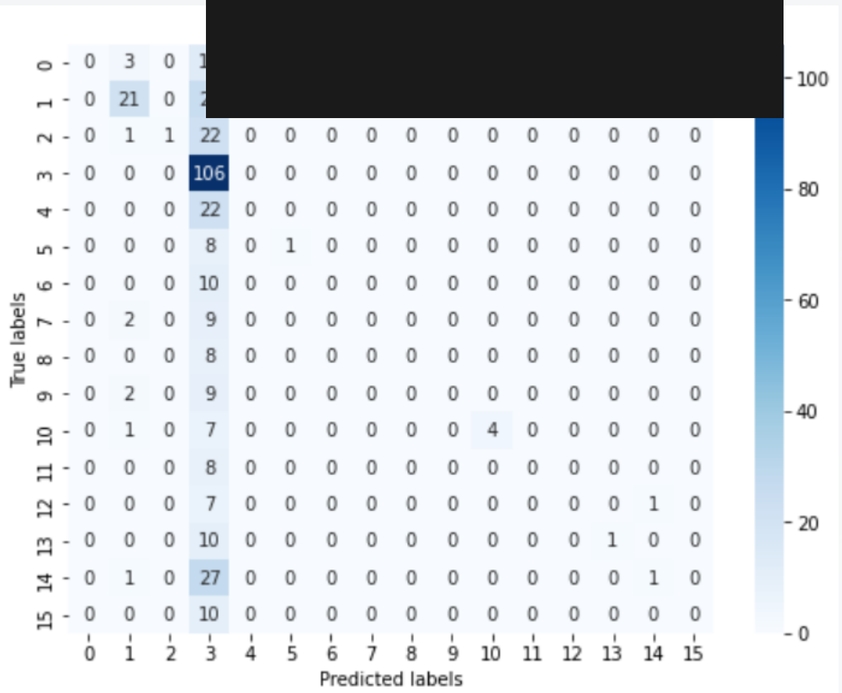
SVM

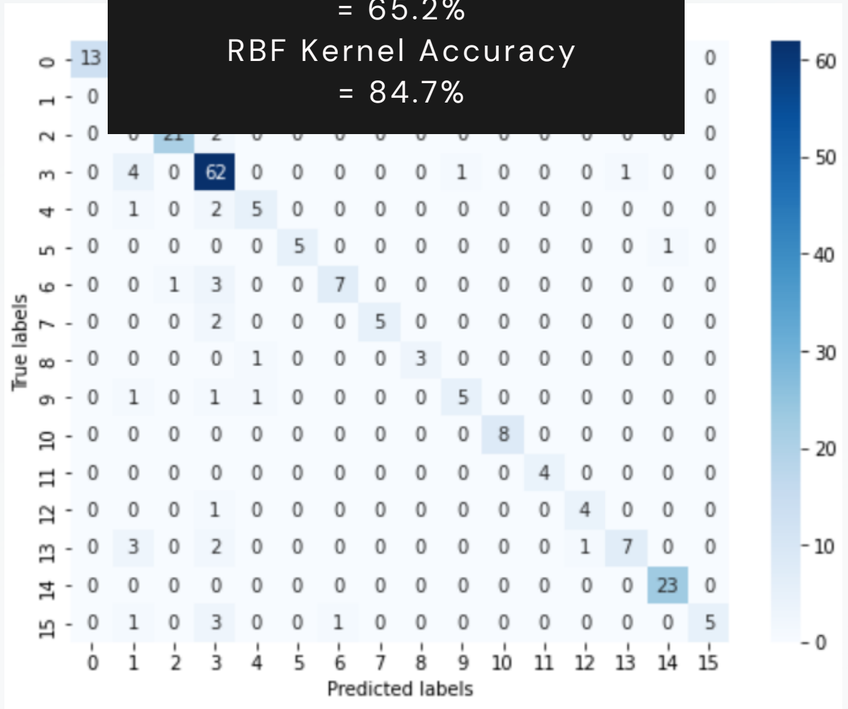
Linear Kernel Accuracy
= 83.2%
Polynomial Kernel Accuracy
= 65.2%
RBF Kernel Accuracy
= 84.7%

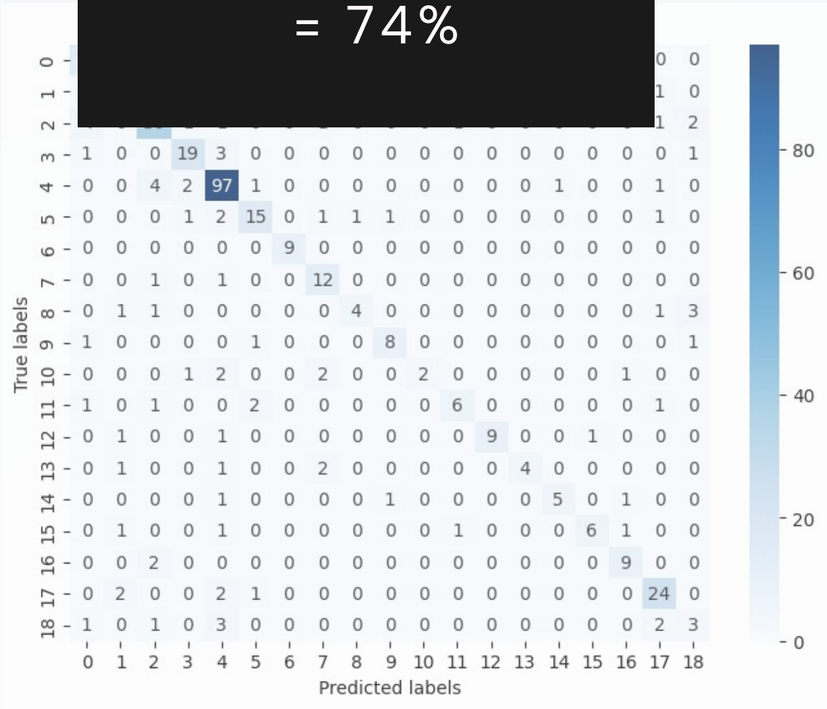


Multilayer Perceptron

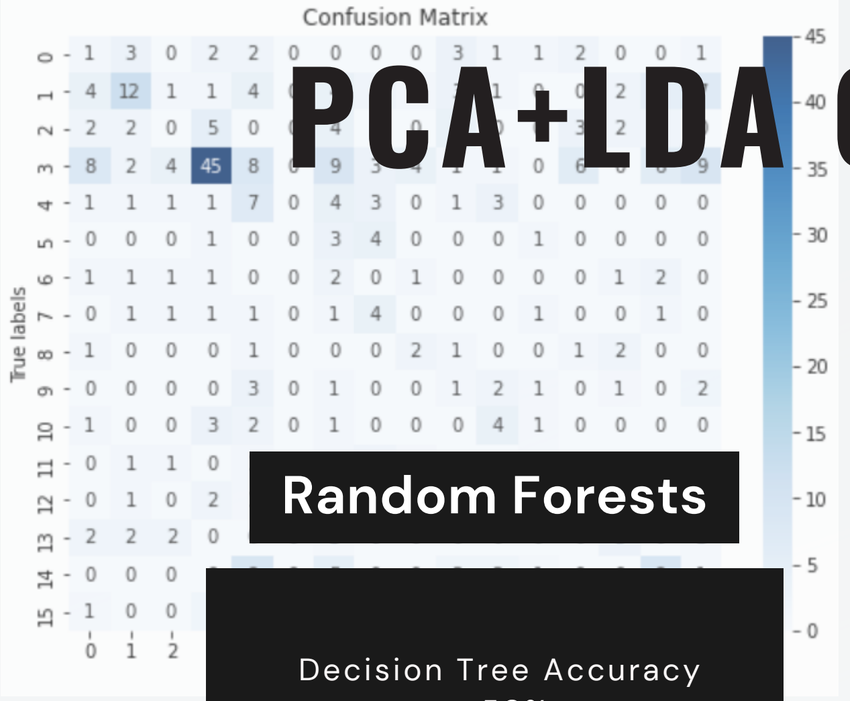
Accuracy
= 74%





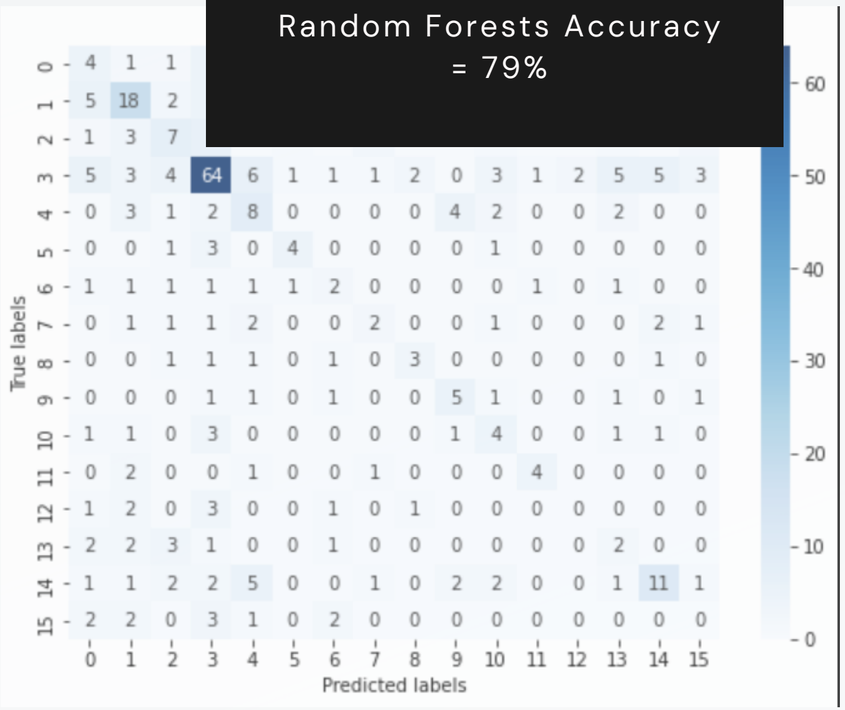


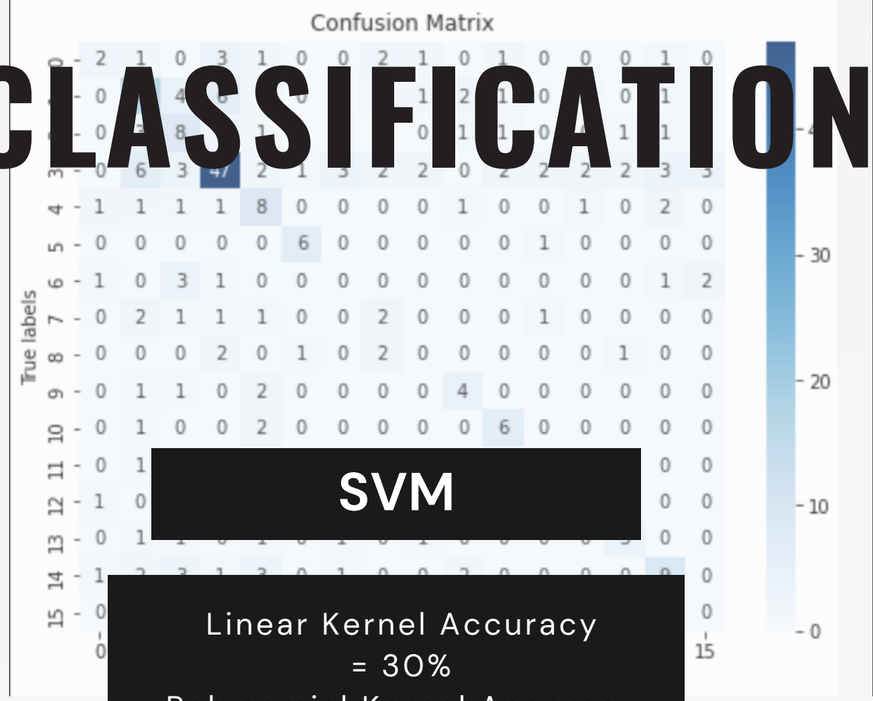
PCA+LDA CLASSIFICATION RESULTS



Random Forests

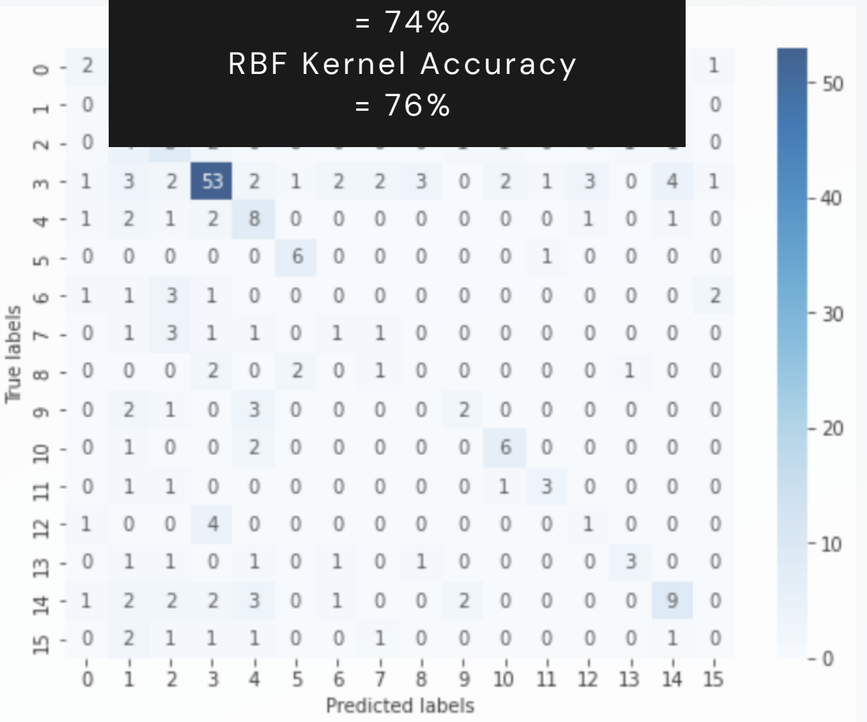
Decision Tree Accuracy
= 58%
Random Forests Accuracy
= 79%

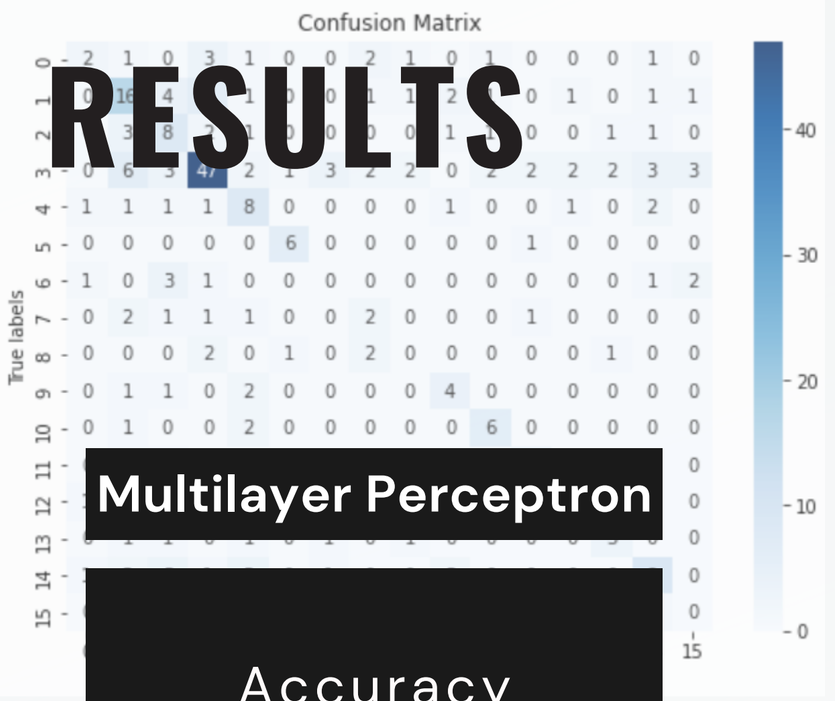




SVM

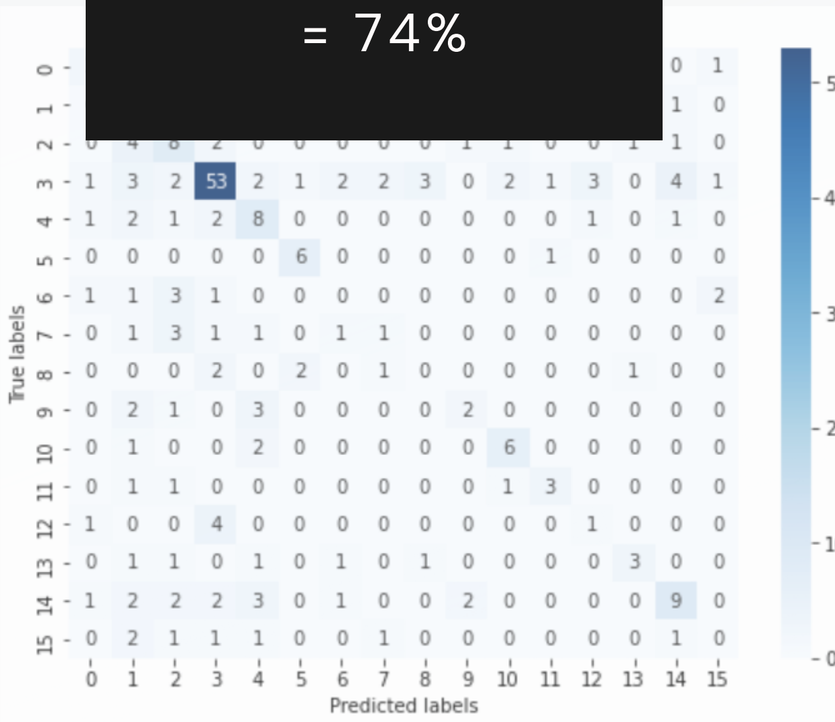
Linear Kernel Accuracy
= 30%
Polynomial Kernel Accuracy
= 74%
RBF Kernel Accuracy
= 76%



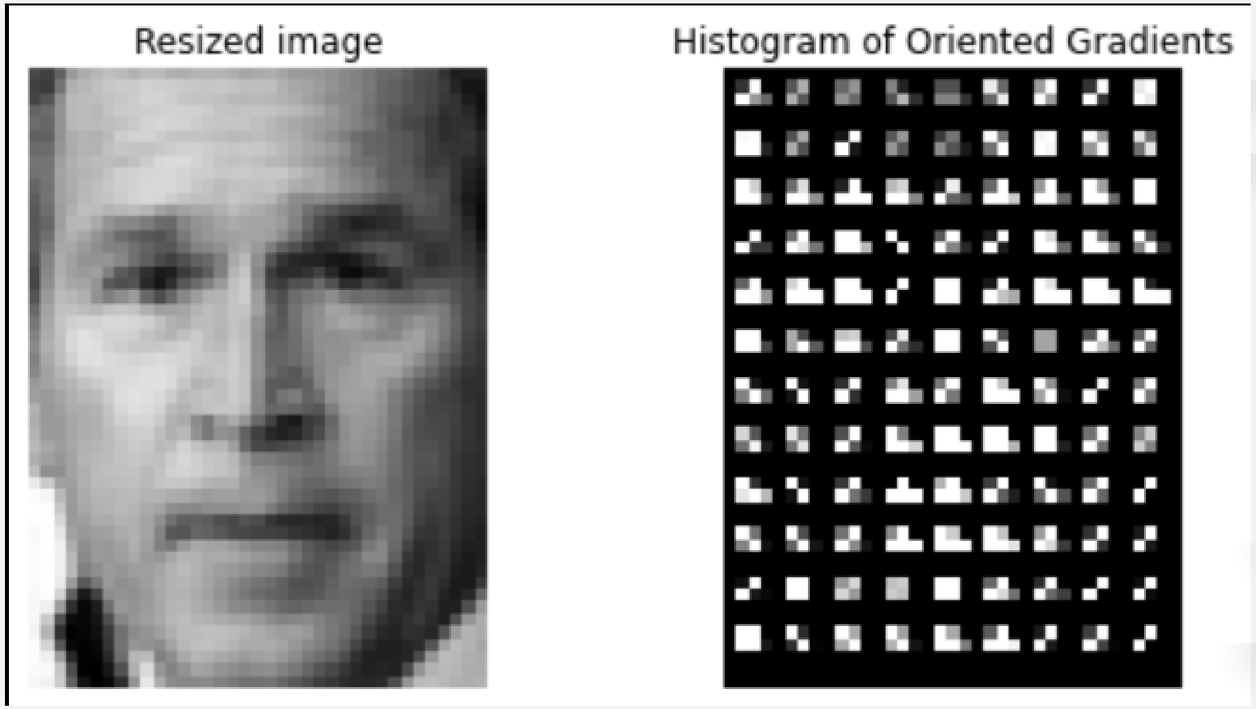


Multilayer Perceptron

Accuracy
= 74%



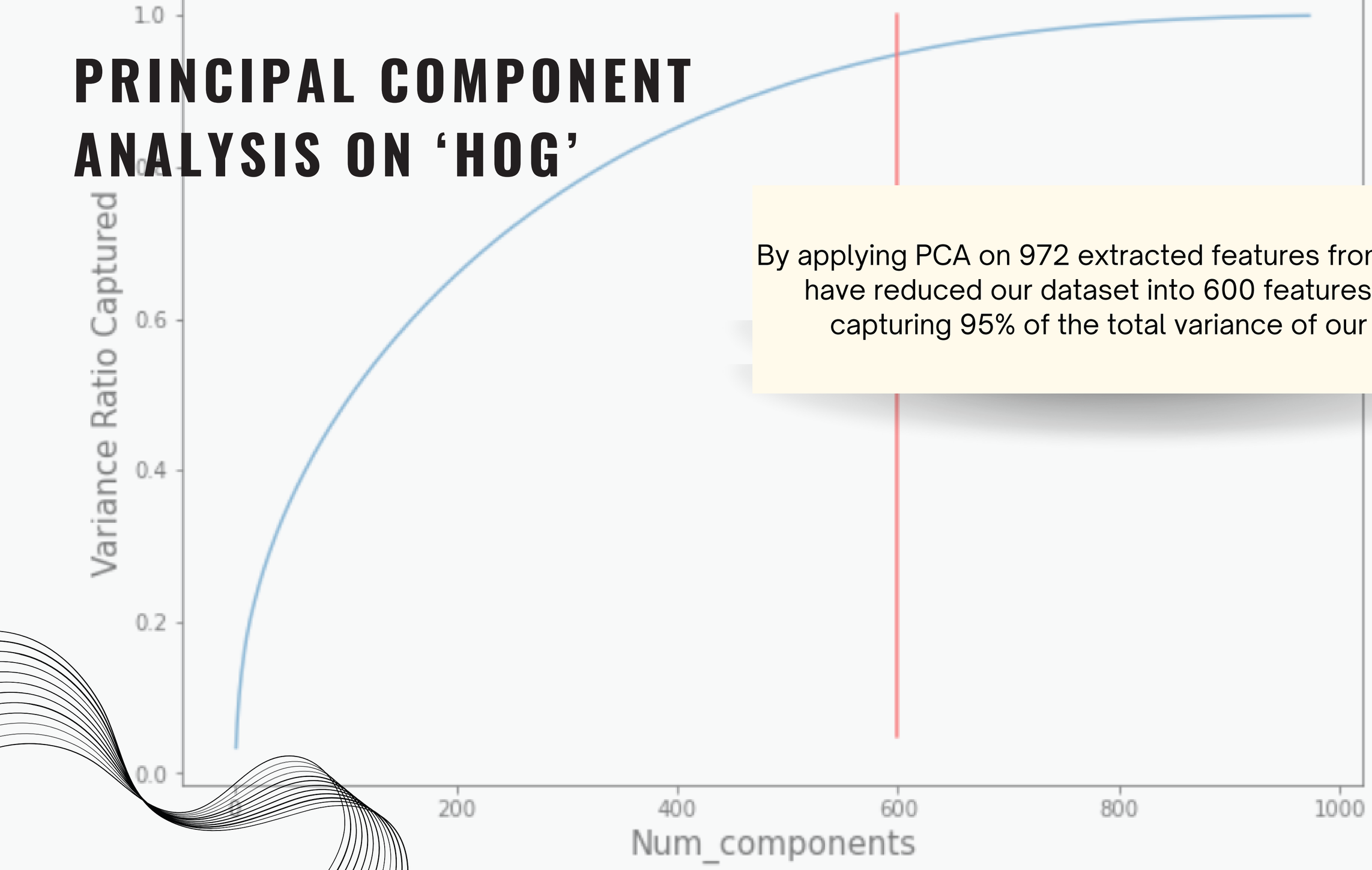
HISTOGRAM OF ORIENTED GRAPHS



HOG extracts local gradient orientation information from an image by dividing it into cells, computing histograms of gradient orientations within each cell, and then normalizing these histograms.

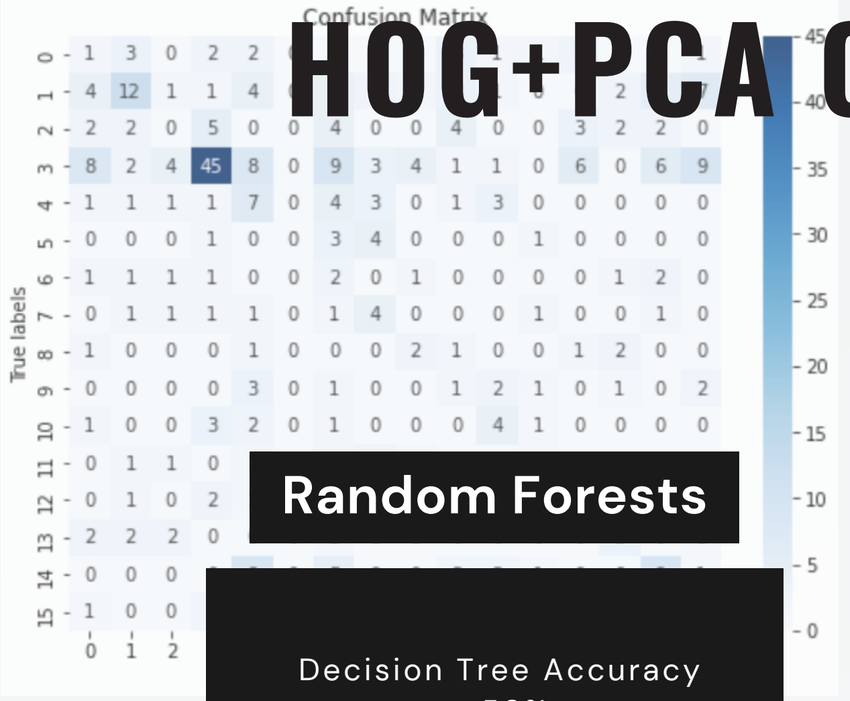
Here each image is divided into cells of (4×4) pixels each and computed 9 HOG orientations for each cell, leading to total of 972 features. Then we have applied techniques like PCA and LDA to further reduce the extracted features for more efficient model training.

PRINCIPAL COMPONENT ANALYSIS ON ‘HOG’



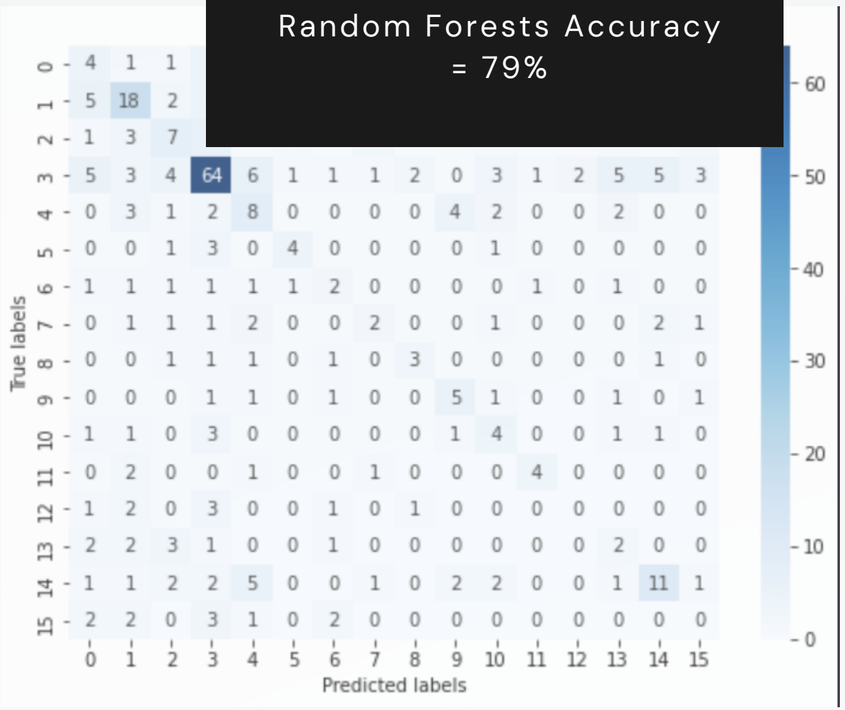
By applying PCA on 972 extracted features from HOG, we have reduced our dataset into 600 features by still capturing 95% of the total variance of our data.

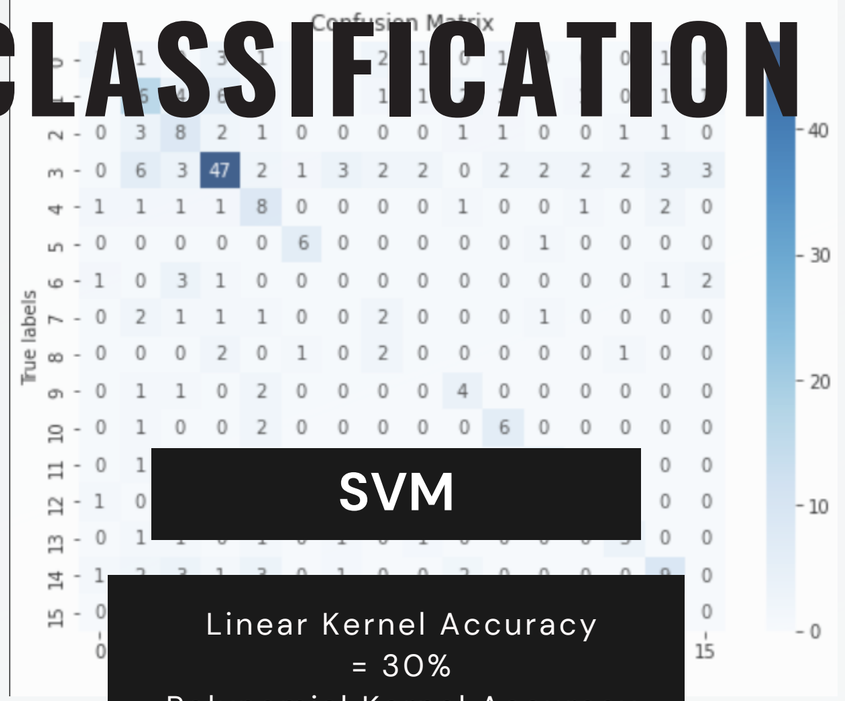
HOG+PCA CLASSIFICATION RESULTS



Random Forests

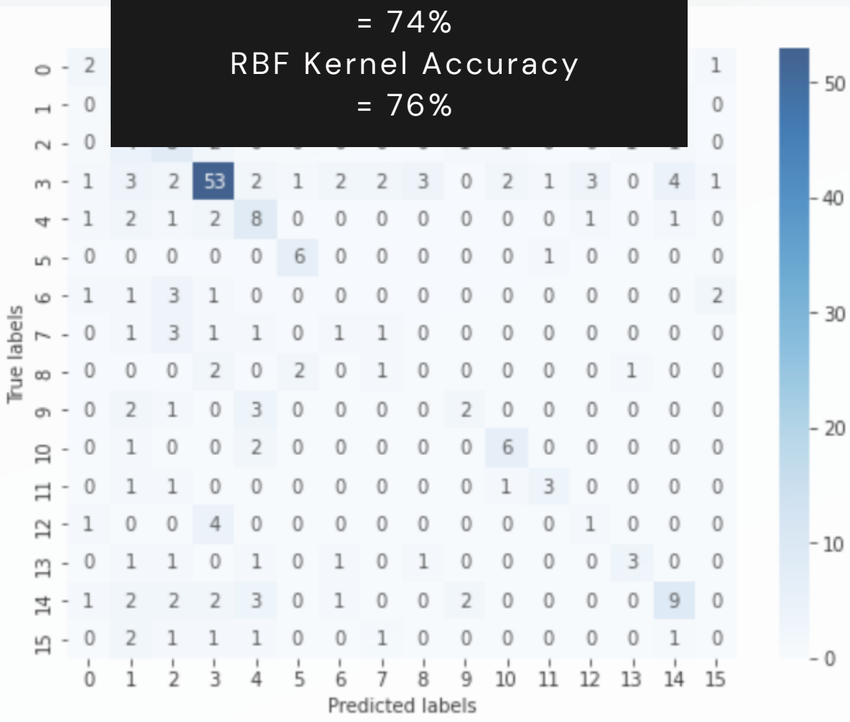
Decision Tree Accuracy
= 58%
Random Forests Accuracy
= 79%

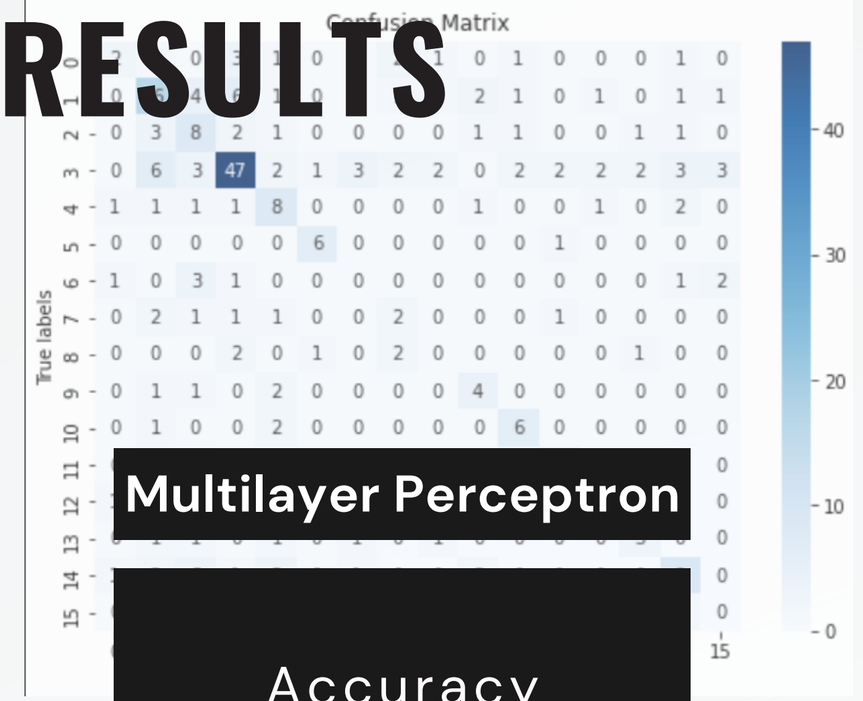




SVM

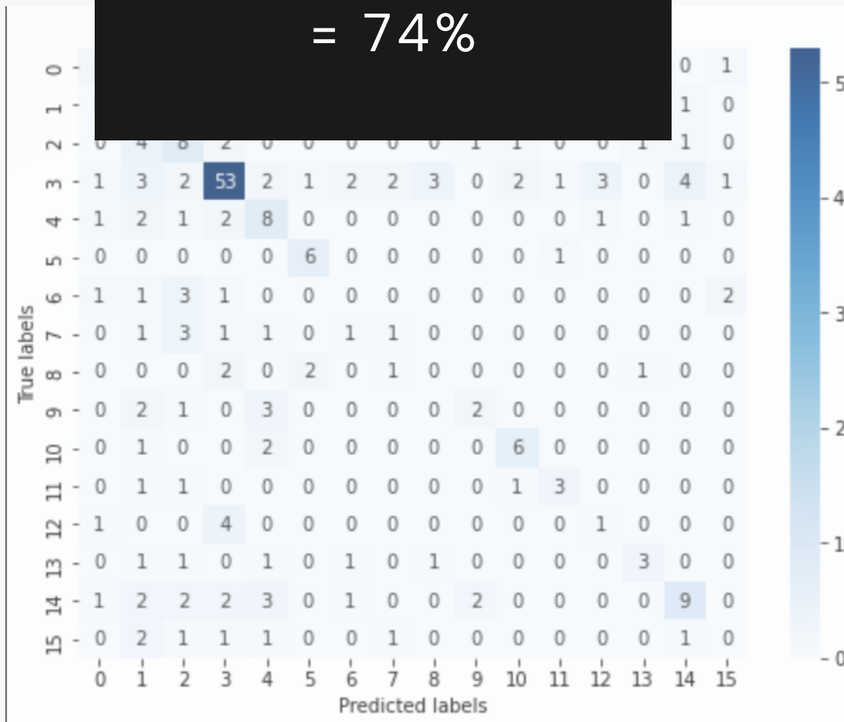
Linear Kernel Accuracy
= 30%
Polynomial Kernel Accuracy
= 74%
RBF Kernel Accuracy
= 76%





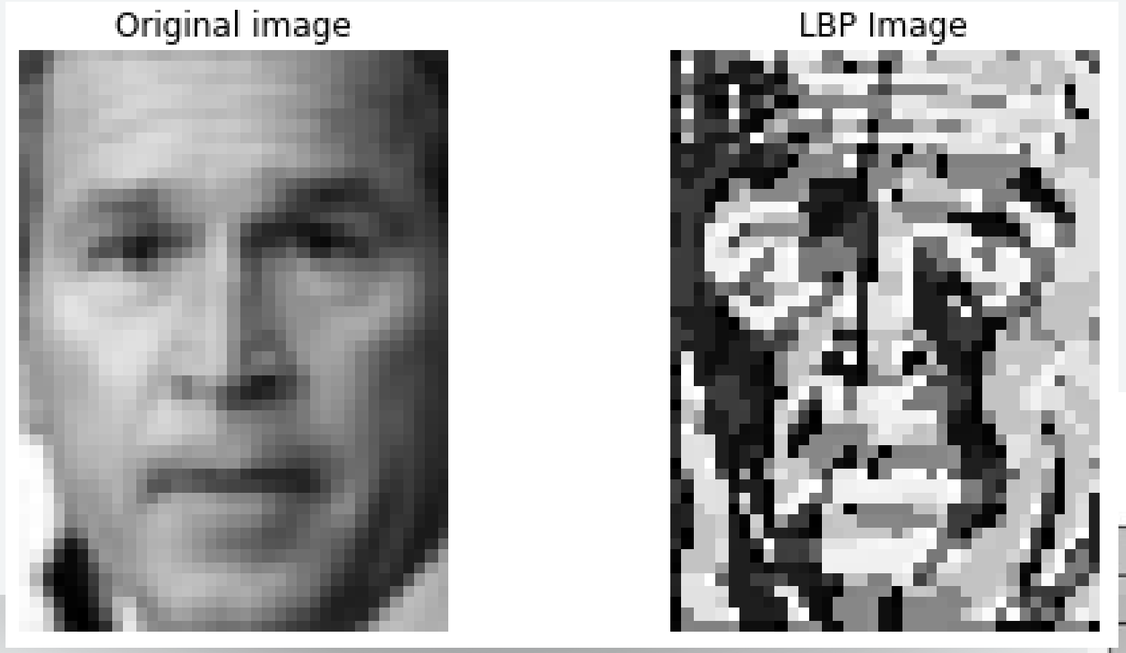
Multilayer Perceptron

Accuracy
= 74%

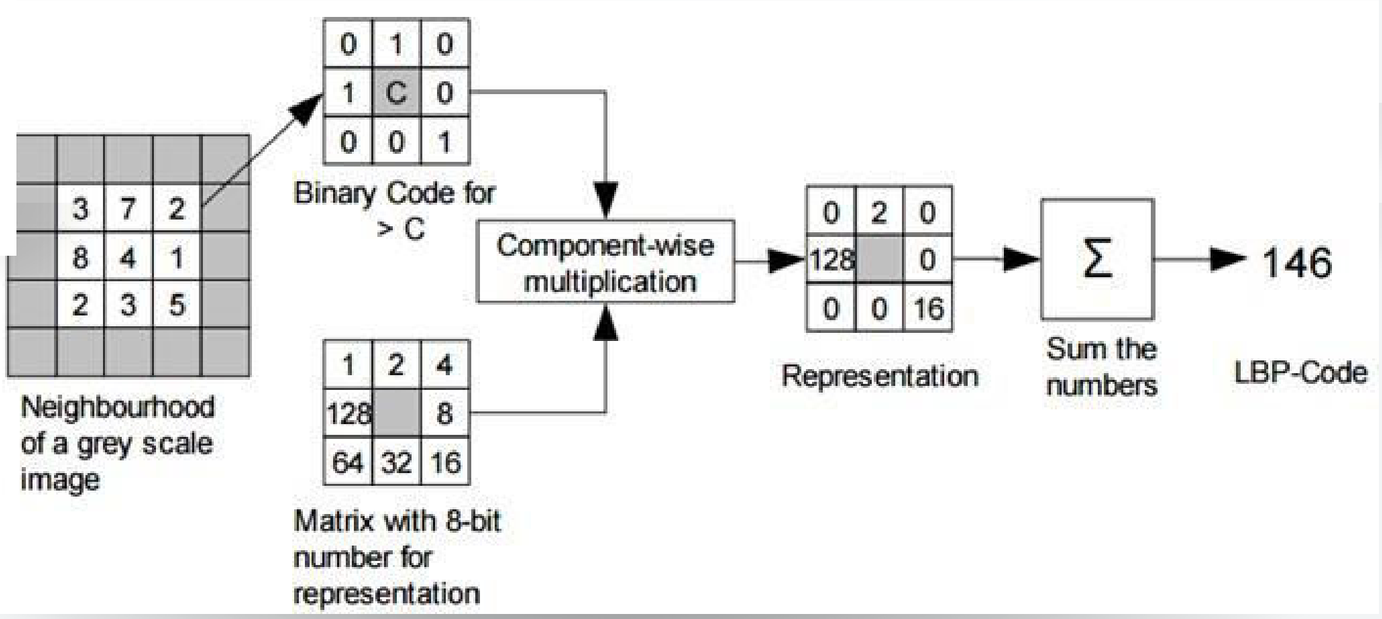


LOCAL BINARY PATTERNS

It characterizes the local texture patterns of an image by comparing the intensity of a pixel with its neighbors.

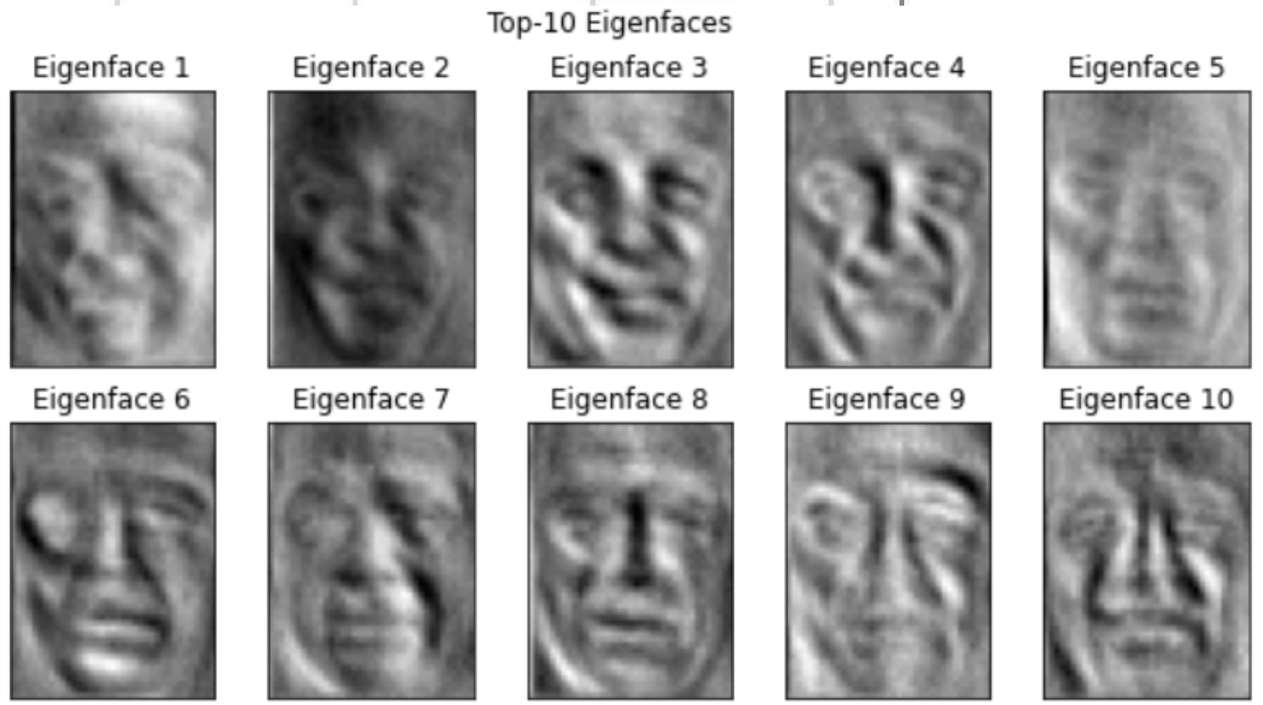


Here we are Calculating new LBP pixel value By Taking its 8 neighbors into account. The difference between the central pixel and its adjacent 8 pixels is extracted as local texture feature representation.

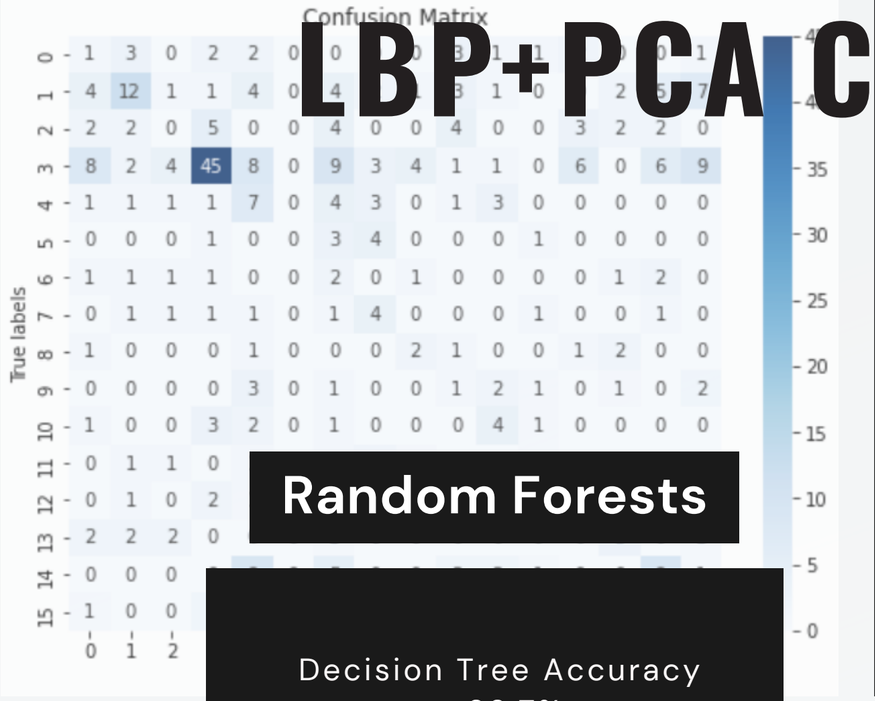


PRINCIPAL COMPONENT ANALYSIS ON ‘LBP’

By applying PCA on 1850 extracted features from LBP, we have reduced our dataset to 600 features and capturing about 90% of Variance and then we apply different Classifiers to these reduced features. LBP is not much effective to face recognition, however it is very effective to identify local textures.

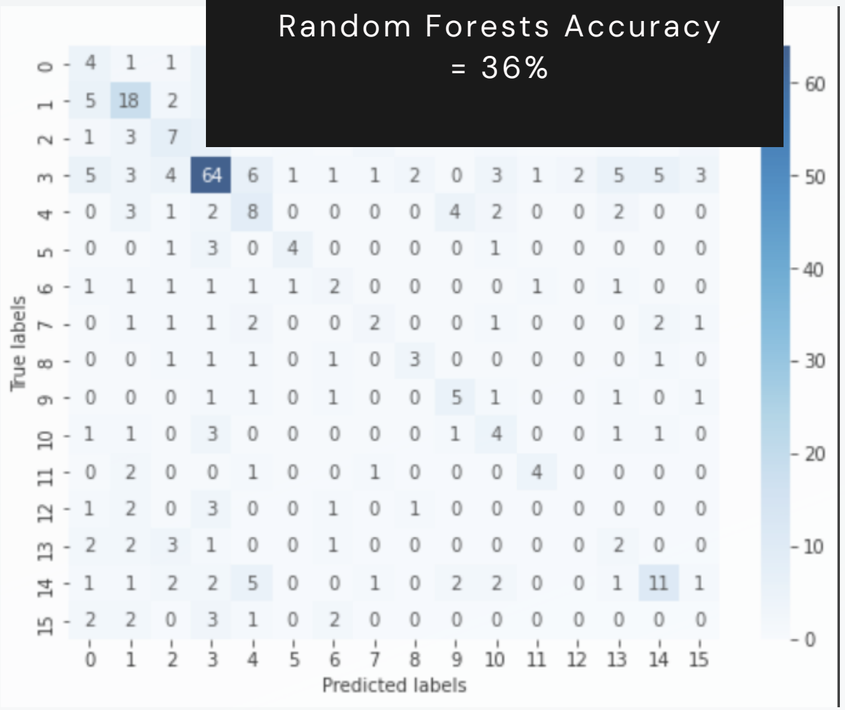


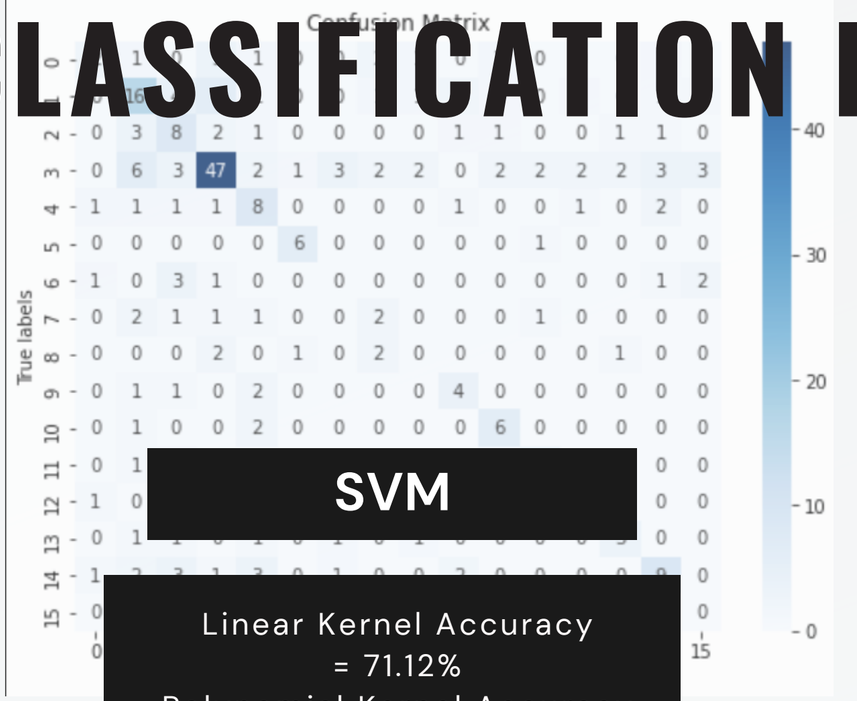
LBP+PCA CLASSIFICATION RESULTS



Random Forests

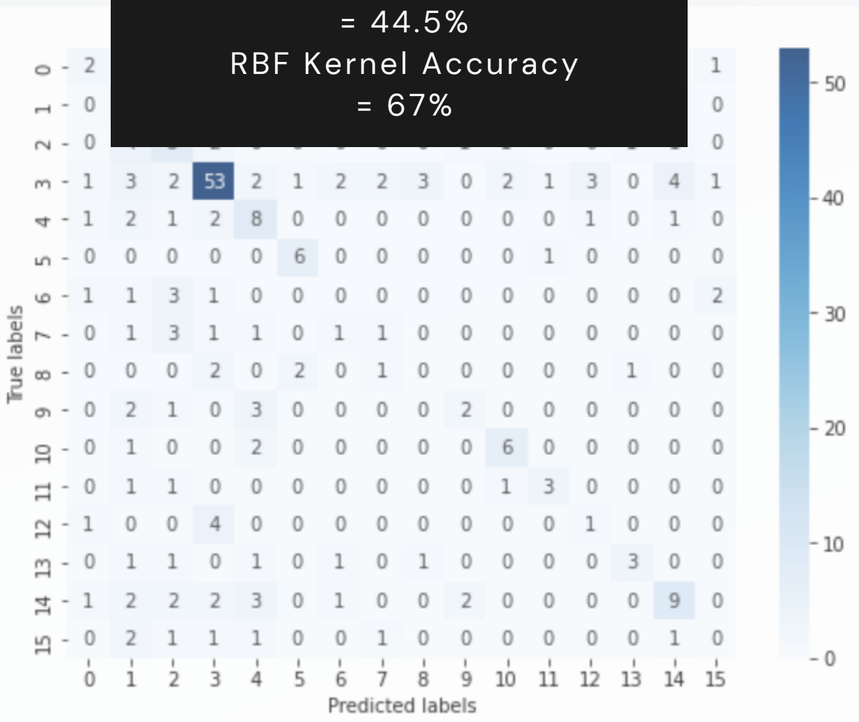
Decision Tree Accuracy
= 33.7%
Random Forests Accuracy
= 36%

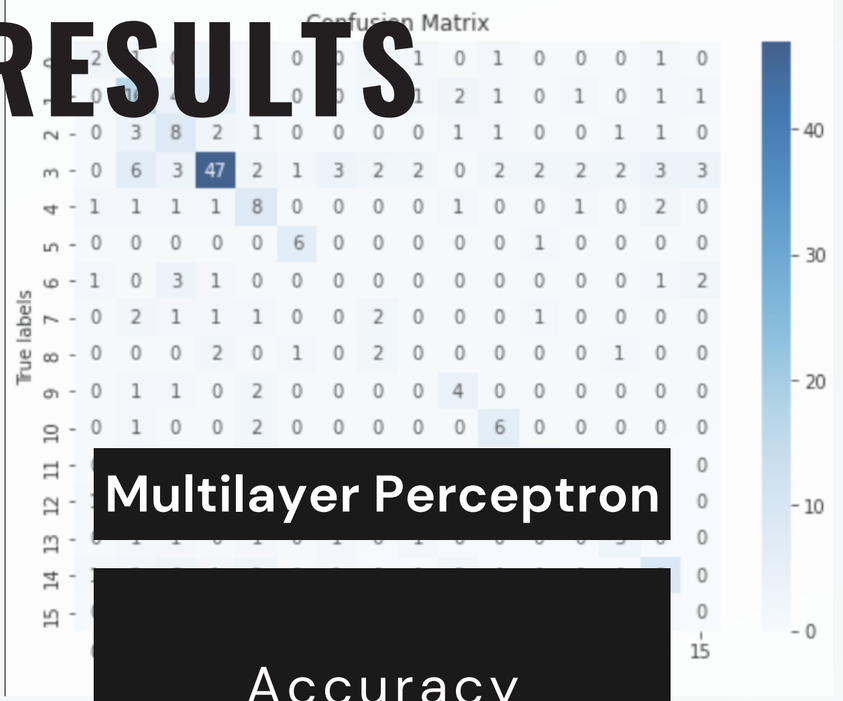




SVM

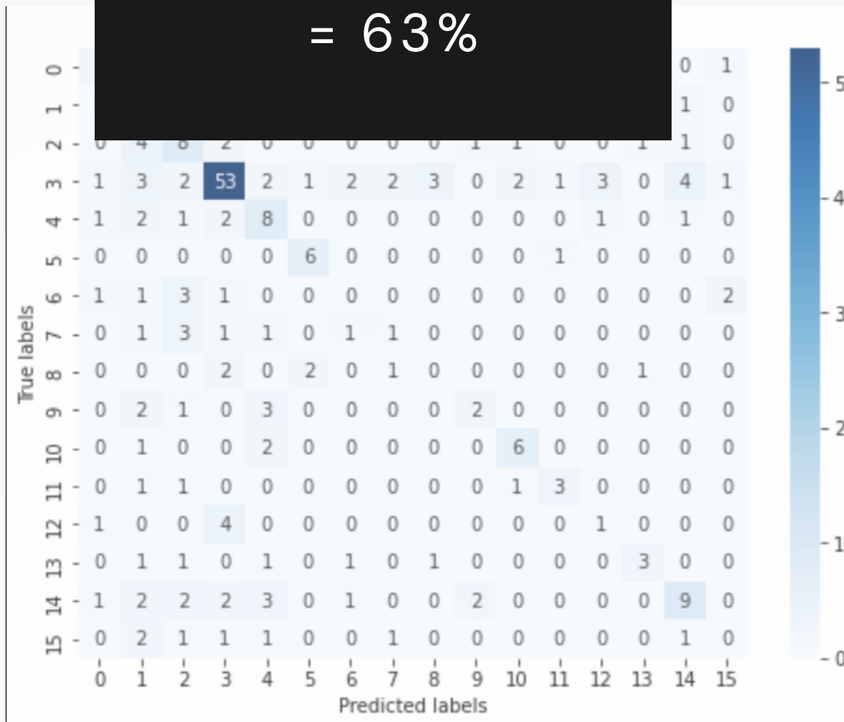
Linear Kernel Accuracy
= 71.12%
Polynomial Kernel Accuracy
= 44.5%
RBF Kernel Accuracy
= 67%



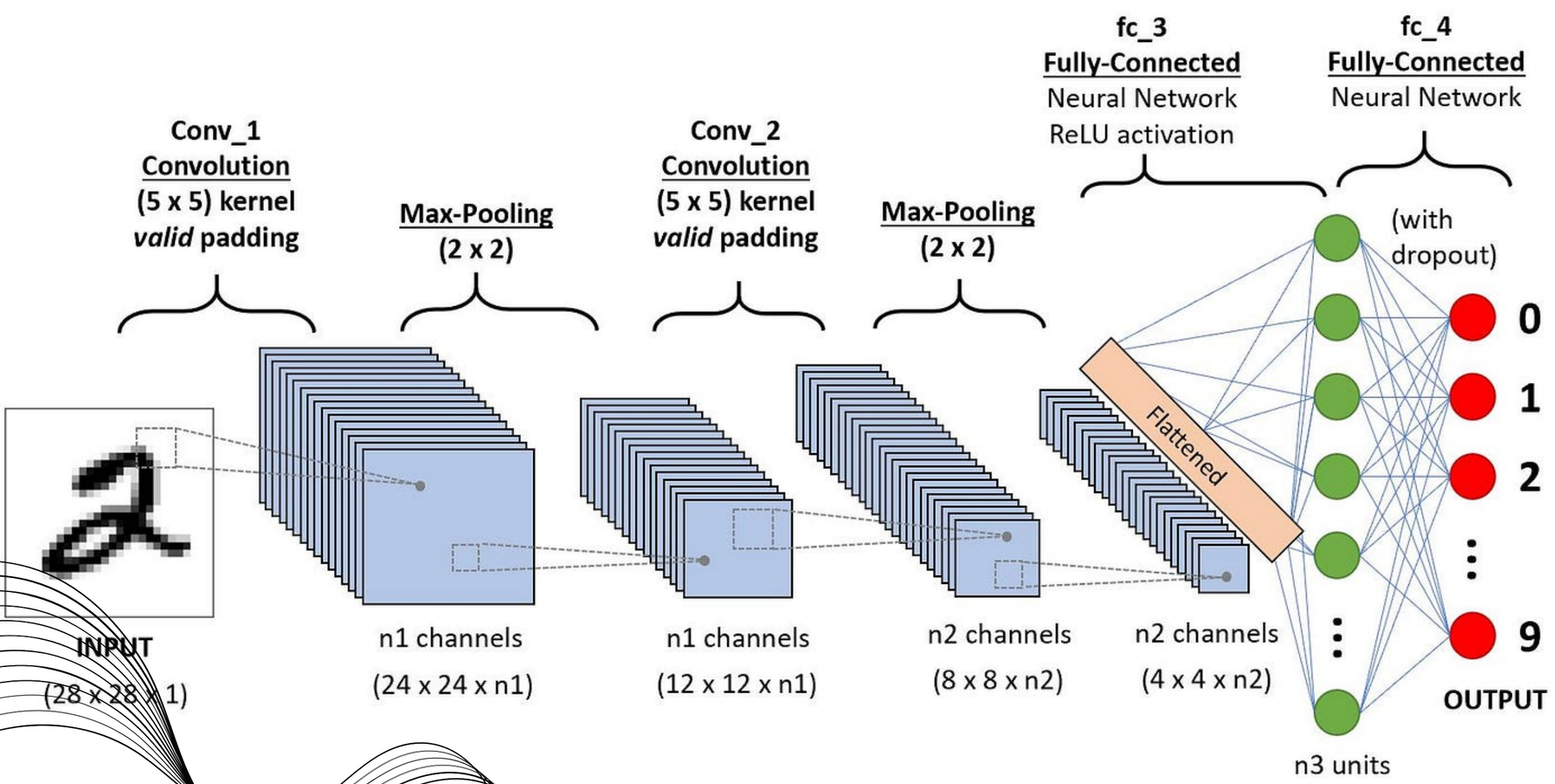


Multilayer Perceptron

Accuracy
= 63%



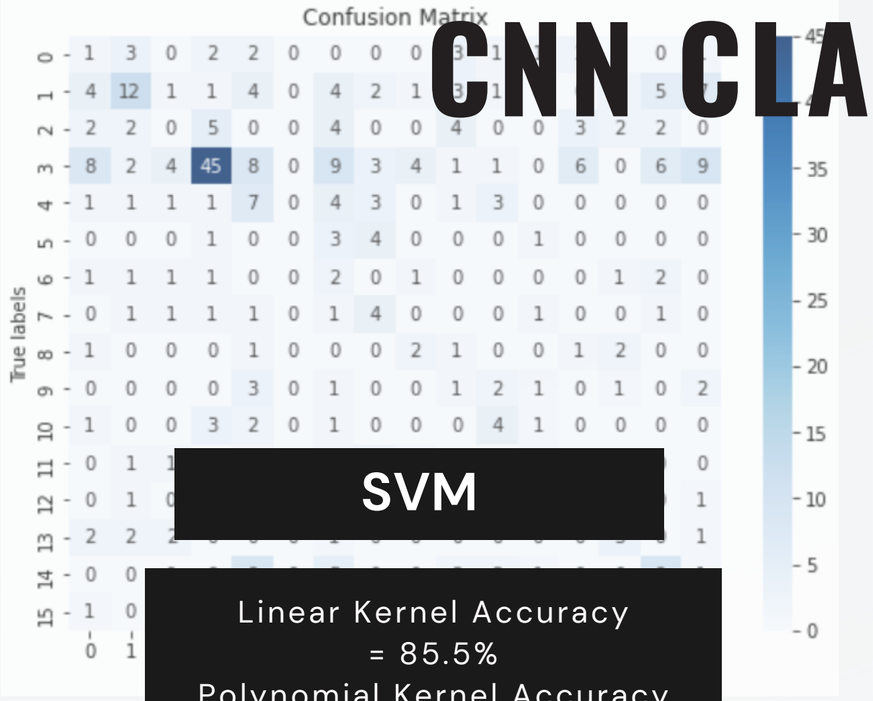
CONVOLUTIONAL NEURAL NETWORKS



To apply CNN on the face data, we have first converted our face images to 60 batches each of batch size 28 samples.

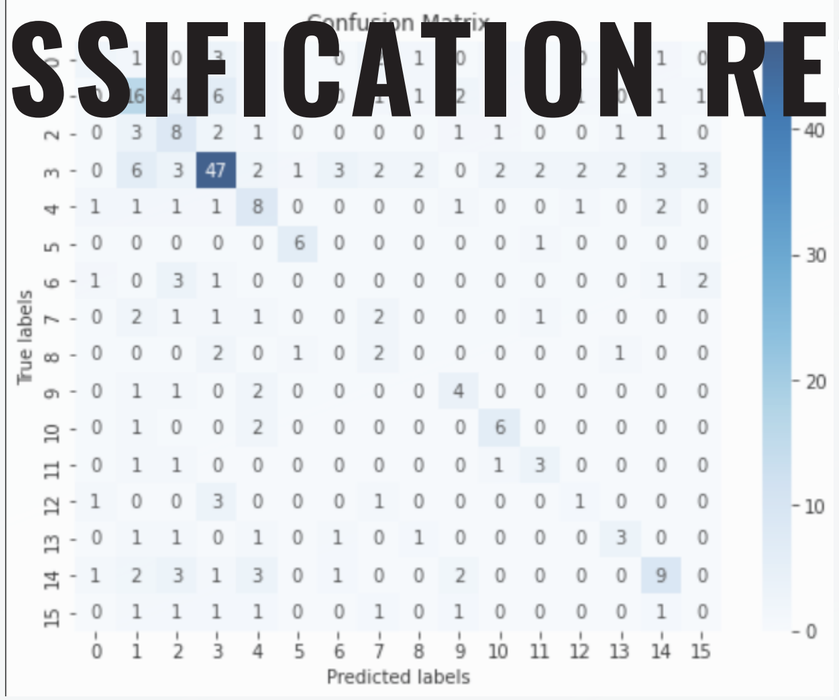
Using CNN, we have have extracted 2048 features. Then we have used classification techniques like SVM, Logistic Regression etc.

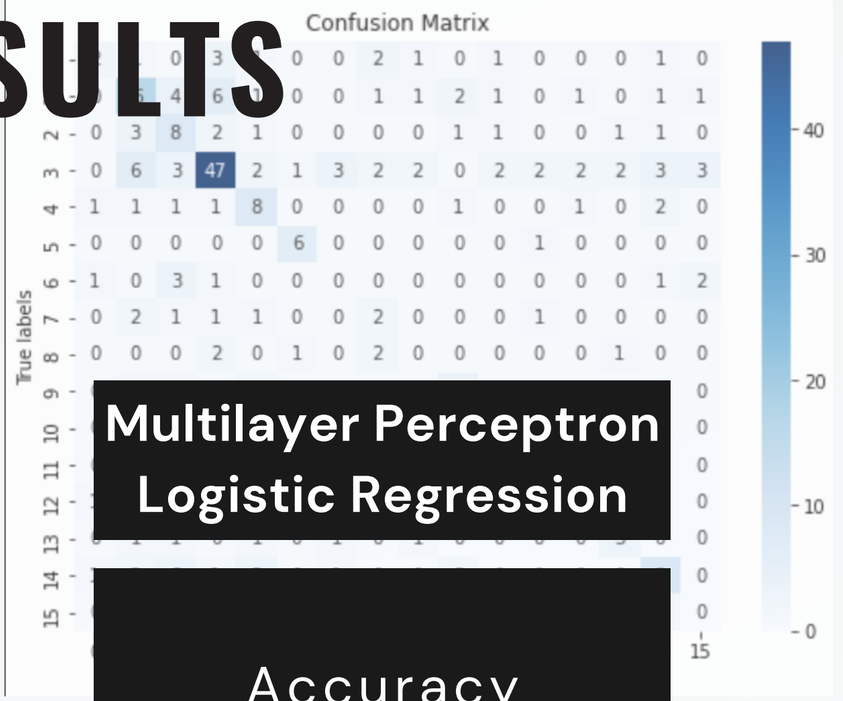
CNN CLASSIFICATION RESULTS



SVM

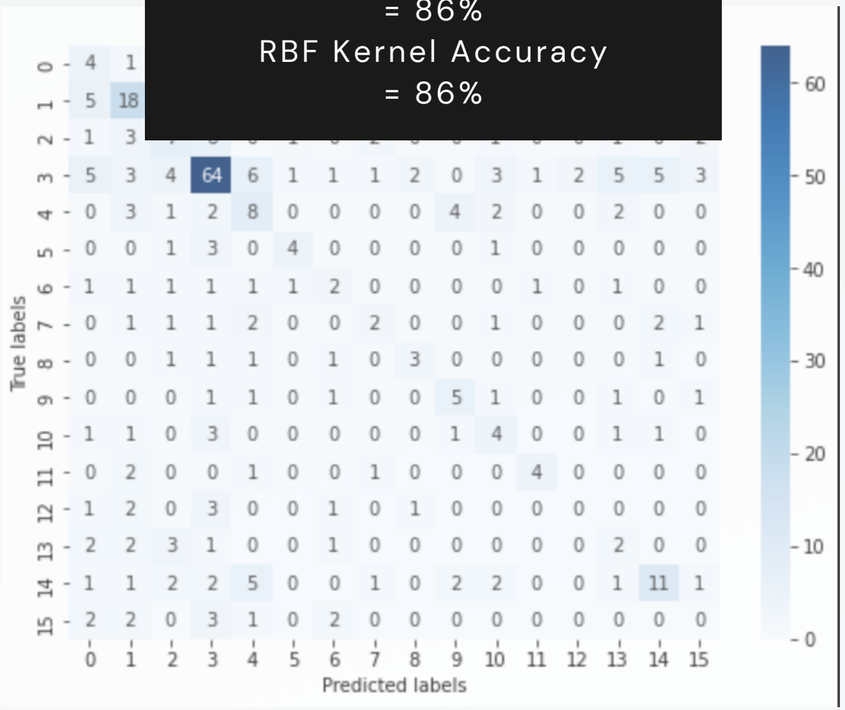
Linear Kernel Accuracy
= 85.5%
Polynomial Kernel Accuracy
= 86%
RBF Kernel Accuracy
= 86%

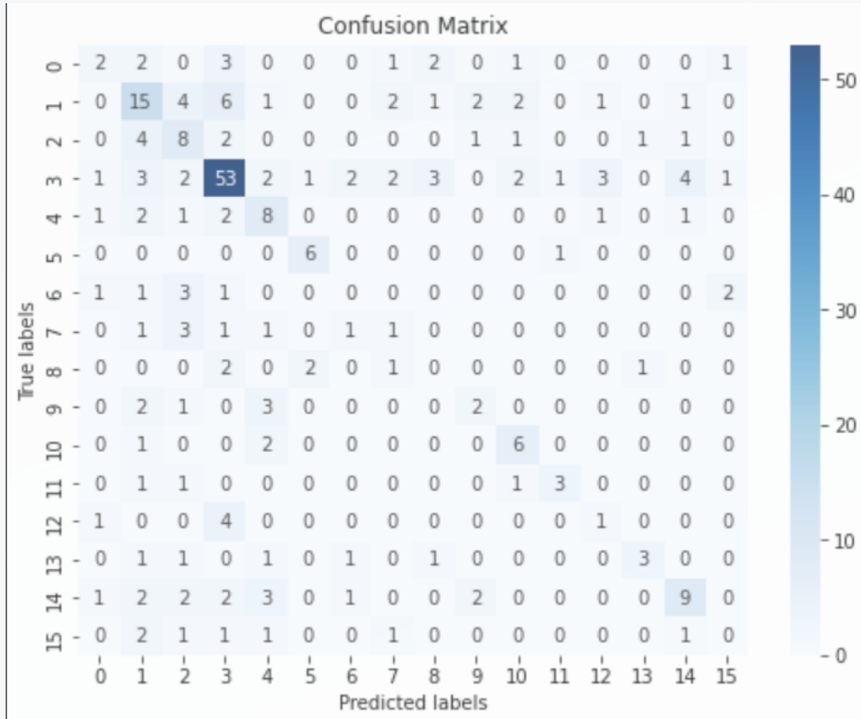


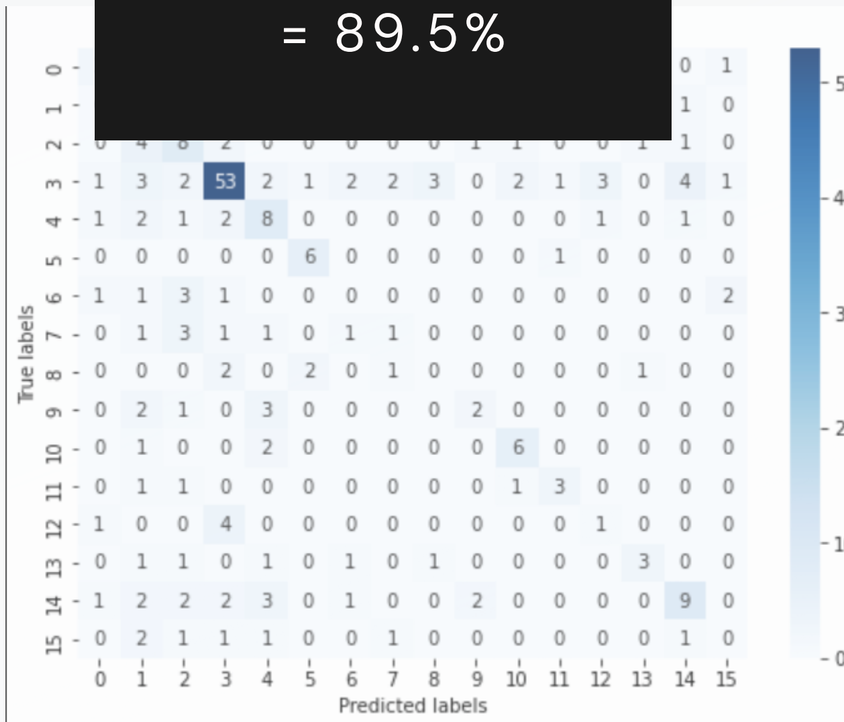


Multilayer Perceptron
Logistic Regression

Accuracy
= 89.5%







FINAL CLASSIFIER SELECTION

01

FEATURE EXTRACTION

HOG

Histogram of Oriented Gradients

02

FEATURE REDUCTION

PCA

Principal Component Analysis

03

CLASSIFIER

RBF SVM

Support Vector Machine

04

ACCURACY

88.38 %

01

FEATURE EXTRACTION

CNN

Convolutional Neural Networks

02

CLASSIFIER

Multilayer Perceptron

03

ACCURACY

89.52 %